Climate Intelligent System

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1 Project Description

1.1 Novel NLP Design

In this project we focus on building an end-to-end NLP pipeline that will automatically extract the structured knowledge from unstructured, scientific climate literature. It organizes this into a Knowledge Graph that is queryable.

The system basically takes the input which are raw scientific data or abstracts. It then identifies the climate-related entities such as CO_2, NO_2, glacier melt, deforestation, sea level rise etc. using **Named Entity Recognition**. It then detects relationships between the entities using **Relation Extraction**. The results are extracted as triples which are then stored as edges and nodes in a **Climate Knowledge Graph**. This knowledge graph will enable graph based analytics like influence ranking, multi-hop impact pathway detection etc.

1.2 What the system is meant to do and what use it will have

The novel approach of the system is in making climate easily accessible, navigable and computable. This project solves the huge problem of a researcher manually reading hundreds of papers to understand how one climate driver influenced the downstream effects. The Climate Intelligent System allows the researchers to query the system and receive a ranked and evidence chain of consequences. The direct applications of the project could be environmental risk assessment and communication of climate science by transforming thousands of research findings into a single interconnected reasoning engine.

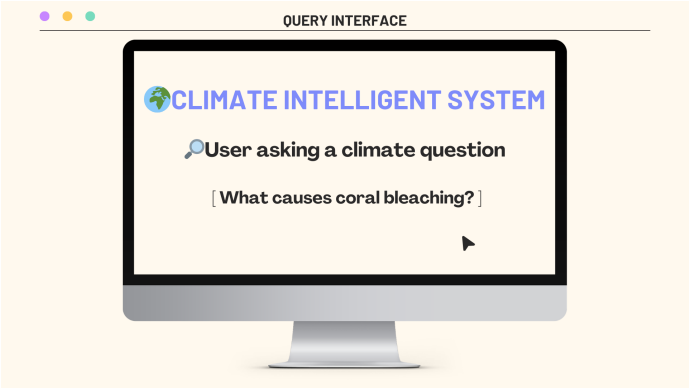

Figure 1: Query interface

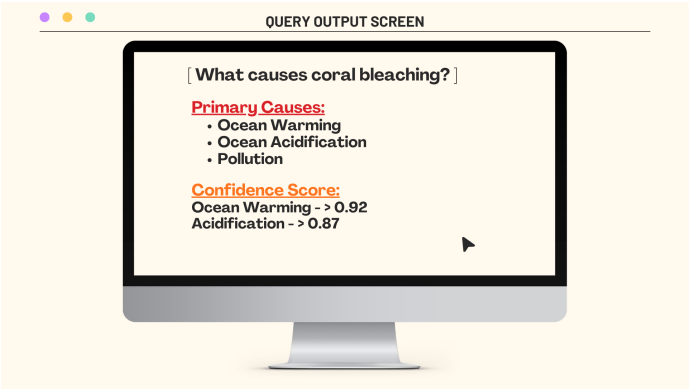

Figure 2: Query output screen

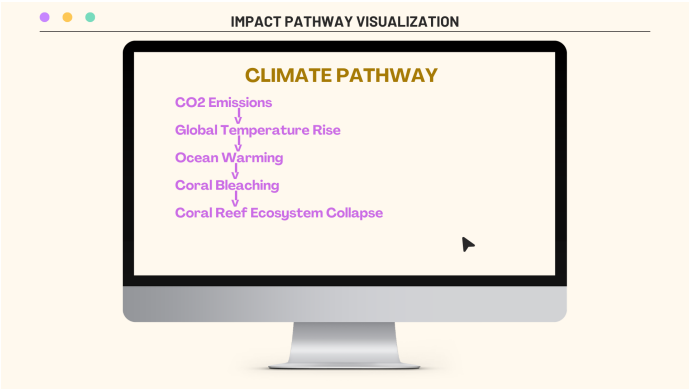

Figure 3: Impact pathway visualization

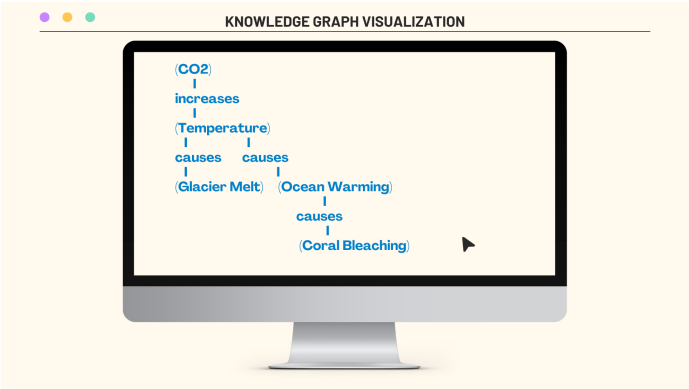

Figure 4: Knowledge graph visualization

Figure 5: Influence ranking dashboard

2 Data

2.1 What type of data is it? What does the data look like (amount, features)? How will you be using the data?

The project would be using scientific climate literature in the form of plain text that consists mainly of primary abstracts and full-text paragraphs. The different data sources that are being considered are:

1. **IPCC Reports** — These are few in number, authoritative, structured climate science documents.
2. **Semantic Scholar API** — They are large-scale scientific abstracts accessible via API.

For our initial plan we want to move ahead with the Semantic Scholar API with approximately 500–1000 climate related abstracts. The API would be queried using the climate-related keywords like CO_2 emissions, global warming, climate change, coral bleaching etc. Each retrieved paper provides the following structured fields:

Table 1: Semantic Scholar API fields used in this project

Field	Usage
`abstract`	Primary text for NER and RE
`title`	Context and filtering
`year`	Filtering for recent literature
`fieldsOfStudy`	Ensuring climate relevance

After annotation each sample will have the following:

- Entity spans labeled across 8 types: Climate Driver, Climate Variable, Environmental Event, Ecosystem, Species, Human Activity, Geographic Location, and Policy.
- Relation triples linking entity pairs, e.g., (CO_2, increases, Temperature).

The data will be used for two purposes:

1. **Training and Evaluation:** A labeled subset of approximately 200–300 annotated sentences will be used to fine-tune both the NER model and the Relation Extraction model, and to evaluate their performance using precision, recall, and F1 score.
2. **Graph Population:** The remaining unannotated abstracts will be processed by the trained pipeline to extract triples at scale, which are then used to construct the Climate Knowledge Graph.

2.2 Prompting strategy

This project does **not** use few-shot or zero-shot prompting. Both the NER and Relation Extraction components are built on fine-tuned transformer models (SciBERT/BERT), trained on labeled climate text. The Query Engine answers user questions purely through graph traversal and ranking over the constructed Knowledge Graph — no LLM prompting is involved at any stage.

3 Proposed Method

3.1 How will you be using algorithms from class to create your system? Will you need to train a model?

This project is built using a multi-staged NLP pipeline. The first stage is **Named Entity Recognition**. The SciBERT model will be fine-tuned to identify and classify climate-related entities from scientific abstracts. SciBERT is chosen over standard BERT because it was pre-trained on a large corpus of scientific text, making it well-suited for domain-specific vocabulary like climate and environmental terminology. The model will be fine-tuned on a labeled dataset of 200–300 annotated sentences

using a token classification objective, where each token is assigned one of 8 entity type labels and a non-entity label. Performance will be evaluated using precision, recall, and F1 score.

The second stage performs **Relation Extraction (RE)**, where another fine-tuned SciBERT model will be trained as a pairwise classifier. Given a sentence and two identified entities, the model predicts the relation between them from a fixed set of 7 relation types: *causes, increases, decreases, affects, contributes_to, occurs_in,* and *mitigates*. The entity pair and surrounding sentence context are fed as input, and the model outputs a relation label, producing structured triples of the form (Entity1, relation, Entity2). These triples form the raw material for the Knowledge Graph.

The third stage constructs and analyzes a **Climate Knowledge Graph** using the extracted triples. Entities become nodes and relations become directed edges, stored and queried using NetworkX. Graph analytics algorithms — specifically Degree Centrality, Betweenness Centrality, and PageRank — are applied to rank the most influential climate drivers and vulnerable ecosystems. Finally, a **Query Engine** is built on top of the graph using graph traversal algorithms to answer user queries such as "What causes coral bleaching?" by finding and ranking connected nodes through their incoming and outgoing edges, enabling multi-hop impact pathway discovery without any LLM prompting.

There will be no few/zero shot prompting the model.

4 Related Work

4.1 What previous research are you basing this idea on?

As this project is a combination of NLP and transformer based model fine-tuning, the following are the related papers that we found are most relevant towards our work.

SciBERT: A Pretrained Language Model for Scientific Text [1] is the most relevant paper to our project. SciBERT is pre-trained on 1.14M scientific papers and 3.1B tokens from Semantic Scholar. We adopt SciBERT as the base model for both our NER and Relation Extraction components, as its scientific pre-training makes it significantly better suited for climate literature than general-purpose BERT.

Towards Dataset for Extracting Relations in the Climate-Change Domain [2] is the closest existing work for our project. It proposes steps to construct a dataset for automatic extraction and structuring of climate-change data using relation extraction and a NER model. It also contains the pipeline design that we aim to implement in our project.

Analyzing Regional Impacts of Climate Change using NLP Techniques [3] demonstrates the feasibility of applying BERT-based NLP to climate literature at scale. It uses BERT-based NER on over 600,000 climate-focused scientific documents to extract location-specific climate entities and conduct region-specific climate trend analyses. We refer to this paper for our approach of using transformer-based NER on climate abstracts.

A Comprehensive Survey on Relation Extraction: Recent Advances and New Frontiers [4] provides the theoretical grounding for our Relation Extraction module. It covers pipeline-based RE approaches that perform entity recognition and relation classification sequentially, as well as deep learning techniques for relation extraction including text representation, context encoding, and triplet prediction.

Hierarchical Transformer Model for Scientific Named Entity Recognition [5] proposes the NER architecture we have adopted for our project. It uses SciBERT as the base encoder combined with a word-level interaction layer and a CRF classification layer, treating NER as BIO tag classification and demonstrating competitive results on scientific datasets.

Scholarly Knowledge Graphs through Structuring Scholarly Communication [6] provides the conceptual basis for our Knowledge Graph construction module. It reviews multiple pipelines for constructing knowledge graphs from scientific literature, including works that fine-tune SciBERT for entity and relation extraction to build structured knowledge graphs from research publications.

5 Team Members

1. **Harshini Domala** - YE32653
   Data collection, preprocessing, and annotation
2. **Ashley Biscoe** - PX70413
   NER model and Relation Extraction model training and evaluation.
3. **Mayukha Sista** - FE67016
   Intelligence Layer - Knowledge Graph construction, Graph Analytics, and Query Engine.

References

[1] Beltagy, I., Lo, K., and Cohan, A. (2019). SciBERT: A pretrained language model for scientific text. Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing *(EMNLP-IJCNLP), 3615–3620.* https://aclanthology.org/D19-1371.pdf

[2] Poleksić, A., and Martinčić-Ipšić, S. (2024). Towards dataset for extracting relations in the climate-change domain. CEUR Workshop Proceedings: TEXT2KG 2024 — Third International Workshop on Knowledge Graph Generation from Text, co-located with ESWC 2024, 3747. https://ceur-ws.org/Vol-3747/text2kg_paper9.pdf

[3] Mallick, T., Murphy, J., Bergerson, J. D., Verner, D. R., Hutchison, J. K., and Levy, L. (2024). Analyzing regional impacts of climate change using natural language processing techniques. arXiv preprint arXiv: 2401.06817. https://arxiv.org/html/2401.06817v1

[4] Zhao, X., Deng, Y., Yang, M., Wang, L., Zhang, R., Cheng, H., Lam, W., Shen, Y., and Xu, R. (2023). A comprehensive survey on relation extraction: Recent advances and new frontiers. arXiv preprint arXiv: 2306.02051. https://arxiv.org/html/2306.02051v3

[5] Zaratiana, U., Holat, P., Tomeh, N., and Charnois, T. (2022). Hierarchical transformer model for scientific named entity recognition. arXiv preprint arXiv: 2203.14710. https://ar5iv.labs.arxiv.org/html/2203.14710

[6] Verma, S., Bhatia, R., Harit, S., and Batish, S. (2022). Scholarly knowledge graphs through structuring scholarly communication: A review. Complex and Intelligent Systems, 9(1), 1059–1095. https://link.springer.com/article/10.1007/s40747-022-00806-6